

1 Aerobic respiration is made up of several stages.

(a) Which is the correct order of the stages of respiration required for the complete oxidation of glucose?

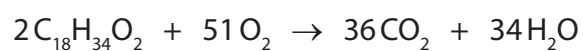
(1)

☐	A	glycolysis	Krebs cycle	link reaction
☐	B	glycolysis	link reaction	Krebs cycle
☐	C	link reaction	Krebs cycle	glycolysis
☐	D	Krebs cycle	glycolysis	link reaction

(b) Write a **balanced** equation for the complete oxidation of glucose in aerobic respiration.

(2)

(c) The equation shows the aerobic respiration of a lipid.



(i) Which is the respiratory quotient when this lipid is used in aerobic respiration?

(1)

- ☐ **A** 0.67
- ☐ **B** 0.71
- ☐ **C** 1.06
- ☐ **D** 1.42

- (ii) The complete oxidation of one molecule of this lipid produces 122 molecules of ATP.

One molecule of this lipid contains 12 244 kJ of energy.

One molecule of ATP yields 30.51 kJ of the energy.

Calculate the percentage of energy in this lipid molecule that can be converted to energy in ATP.

Give your answer to **three** significant figures.

(2)

Answer %

(Total for Question 1 = 6 marks)